

JK Flip-Flop Symbol and Truth Table

- In the [SR Flip-Flop](#), when both inputs S and R are 1 then the output of the flip-flop is indeterminate. That issue can be resolved using the JK Flip-Flop.
- Similar to the SR Flip-Flop, the JK flip-flop has two inputs. And using the two inputs the flip-flop can be set, reset, hold (memory) or toggled.
- Unlike the SR flip-flop, in the JK flip-flop, when both inputs J and K are 1 then the output of the flip-flop toggles.

Symbol

Truth Table

CLK	J	K	Q_{n+1}
↑	0	0	Q_n
↑	0	1	0
↑	1	0	1
↑	1	1	Q_n'

- The symbol of edge triggered JK flip-flop and its truth table is shown below.

JK Flip-Flop Symbol and Truth Table

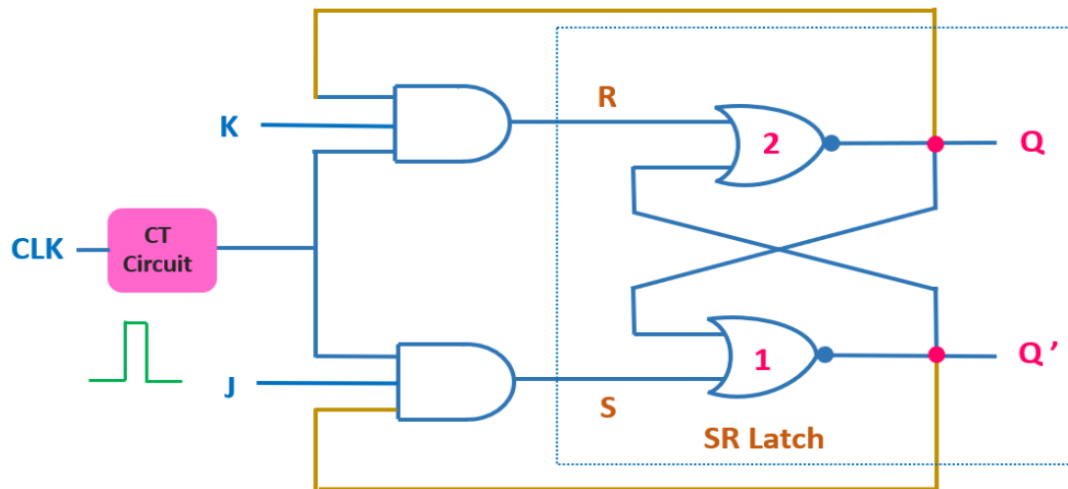
- In the JK flip-flop, at the rising edge of the clock,
 When **J = 0 and K = 0** then flip-flop **retains (holds) the current state**.
 When **J = 0 and K = 1**, then flip-flop **resets to 0**.
 When **J = 1 and K = 0**, then flip-flop **sets the output to 1**.
 And when **J = 1 and K = 1** then **output of the flip-flop toggles**.
- When the clock signal is low, then irrespective of the value of J and K inputs, the flop-flop retains the present state.
- The detailed truth table with all different possibilities of Q_n , J and K inputs are shown below, where Q_n is the present state and Q_{n+1} is the next state of the flip-flop.

CLK	J	K	Q_n	Q_{n+1}	Q_{n+1}
0	X	X	0/1	0/1	Q_n
↑	0	0	0	0	Q_n
	0	0	1	1	
↑	0	1	0	0	0
	0	1	1	0	
↑	1	0	0	1	1
	1	0	1	1	
↑	1	1	0	1	Q_n'
	1	1	1	0	

JK Flip-Flop Truth Table

JK Flip-Flop Circuit Diagram

With the little modification in the circuit of the SR flip-flop circuit, it can be used as the JK flip-flop. The circuit of JK flip-flop is shown below.



Circuit Diagram of Positive Edge Triggered JK Flip-Flop

As you can see from the circuit diagram, in the RS flip-flop circuit, the K input is applied in place of the R input while the J input is applied in place of the S input. Moreover, there is a feedback from output to input side. The Q output is connected back the AND gate where the K input is applied while the Q' output is connected back to the AND gate where the J input is applied. Here the clock transition circuit generates the narrow pulses at the every clock transition. The same is applied to the enable input of the latch. And in this way, the gated latch behaves as an edge triggered flip-flop.

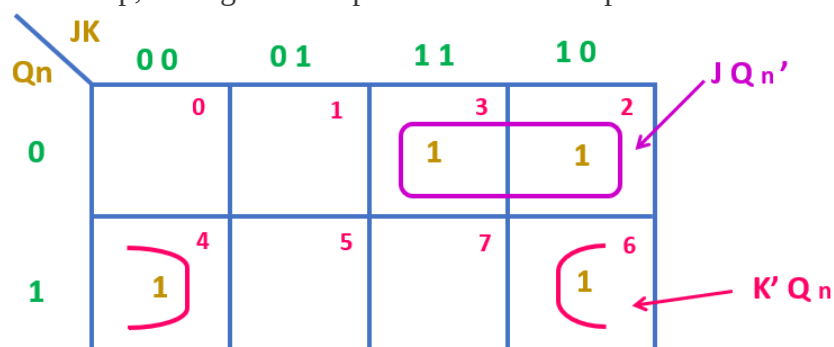
JK Flip-Flop Characteristic Equation

The characteristic equation shows the output of the flip-flop Q_{n+1} in terms of the present state Q_n and the current inputs J and K. The characteristic table of the JK flip-flop is shown below.

	Q_n	J	K	Q_{n+1}
m4	0	0	0	0
	1	0	0	1
	0	0	1	0
	1	0	1	0
m2	0	1	0	1
m6	1	1	0	1
m3	0	1	1	1
	1	1	1	0

The characteristic Table of JK Flip-Flop

As per the characteristic equation, the output Q_{n+1} is '1' for 4 different input combinations. (4 minterms). Using the K-map, the algebraic expression can be simplified further.



As per the K-map, after the simplification the output of the flip-flop or the characteristic equation

$$Q_{n+1} = J Q_n' + K' Q_n$$

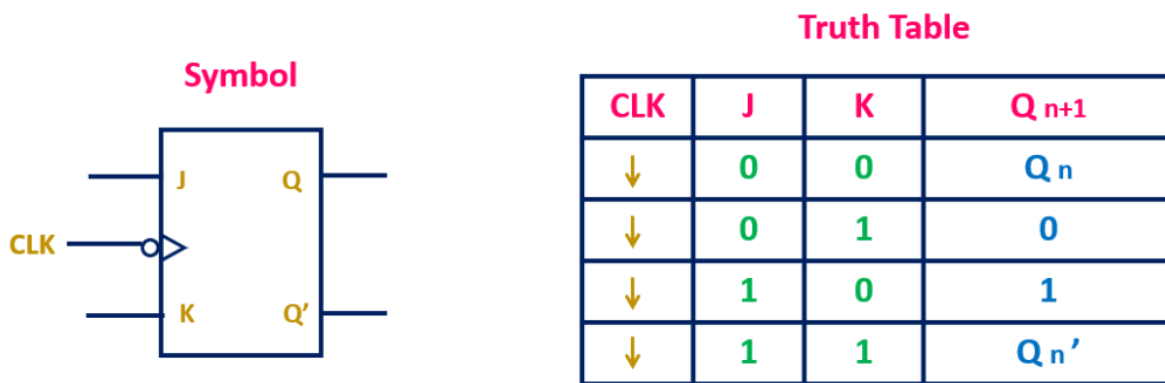
JK Flip-Flop Excitation Table

The excitation table of the flip-flop shows the required excitation to the flip-flop, or the required input to the flip-flop, to go from the given state to the next particular state. The excitation table of the JK flip-flop is shown below. In the table, 'X' represents that the value of input variable can be either '0' or '1'.

Q_n	Q_{n+1}	J	K
0 → 0	0	0	X
1 → 0	0	X	1
0 → 1	1	1	X
1 → 1	1	X	0

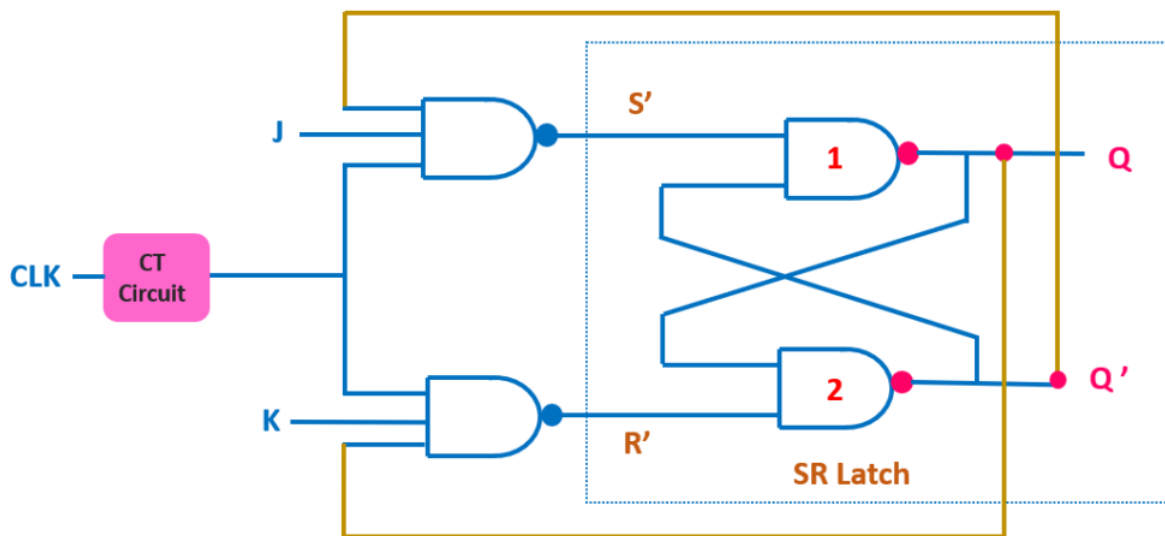
Excitation Table of JK Flip-Flop

The negative edge triggered JK flip-flop is similar to the positive edge triggered flip-flop. But it responds to the inputs only at the falling edge of the clock. The symbol and truth table of negative edge triggered flip-flop is shown below.



Symbol and Truth Table of Negative Edge Triggered JK Flip-Flop

The same circuit can also be implemented using the NAND gates. The circuit diagram of the JK flip-flop with the NAND gates is shown below.



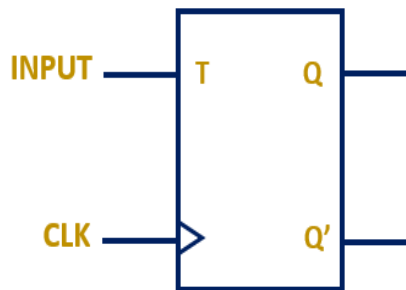
Circuit Diagram of Positive Edge Triggered JK Flip-Flop (using NAND gates)

T Flip-Flop Explained | Working, Circuit diagram, Excitation Table and Characteristic Equation of T Flip-Flop

T Flip-Flop Symbol and Truth Table

The T flip-Flop has one input.

- When the input is 1 then its output toggles. Let's say the present state of the flip-flop is Q_n . So, with $T = 1$, if $Q_n = 0$ then in the next state, the output of the flip-flop Q_{n+1} will become 1. And similarly currently if Q_n is 1 then in the next state, the output will become 0.
- With $T = 0$ input, since the output of the flip-flop **toggles** at every clock pulse, so it is known as T flip-flop. When $T = 0$, then the flip-flop will remain in the same state. The symbol and the truth table of the T flip-flop is shown below.

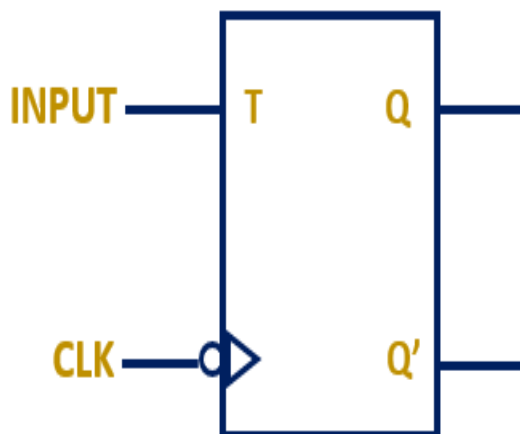


Symbol of Positive Edge Triggered T Flip-Flop

CLK	T	Q_{n+1}
↑	0	Q_n
↑	1	Q_n'

Truth Table of Positive Edge Triggered T Flip-Flop

The symbol which is shown above is the symbol of the positive edge triggered flip-flop. The triangle in the symbol indicates that, the flip-flop responds to the input only at the rising edge of the clock. Similarly, the symbol of the negative edge triggered T flip-flop is shown below.



Symbol of Negative Edge Triggered T Flip-Flop

CLK	T	Q_n	Q_{n+1}
↓	0	0	0
↓	0	1	1
↓	1	0	1
↓	1	1	0

Truth Table of Negative Edge Triggered T Flip-Flop

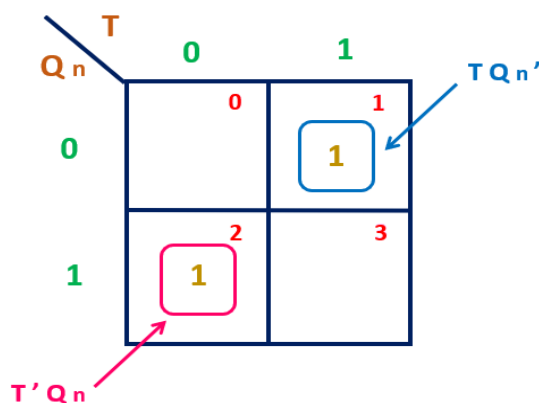
Characteristic Equation of T Flip-Flop

The characteristic equation of the flip-flop is the algebraic representation of the next state of the Flip-Flop (Q_{n+1}) in terms of the present state (Q_n) and the current input (T).

That means, here the input variables are Q_n and T, while the output is Q_{n+1} .

Q_n	T	Q_{n+1}
0	0	0
1	0	1
0	1	1
1	1	0

Boolean expression.



$$Q_{n+1} = TQ_n' + T'Q_n$$

$$= T \oplus Q_n$$

Excitation Table of T Flip-Flop

The excitation table of the flip-flop shows the required excitation to the flip-flop, or the required input to the flip-flop, to go from the given state to the next particular state. For the T input, when $T = 0$ then the flip-flop remains in the same state and when $T = 1$ then the output of the flip-flop toggles at every clock pulse. The excitation table of the T flip-flop is shown below.

Q_n	Q_{n+1}	T
0 → 0	0	0
1 → 0	0	1
0 → 1	1	1
1 → 1	1	0

Circuit Diagram of T Flip-Flop

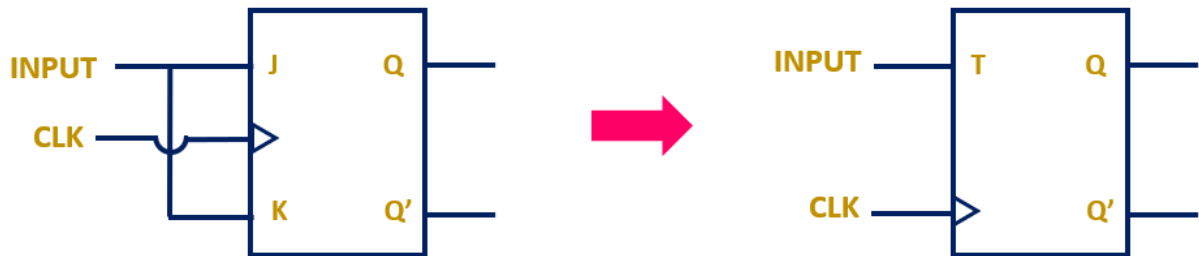
With the little modification, the JK flip-flop can be used as a T flip-flop. The truth table of the T flip-flop is shown below.

Symbol

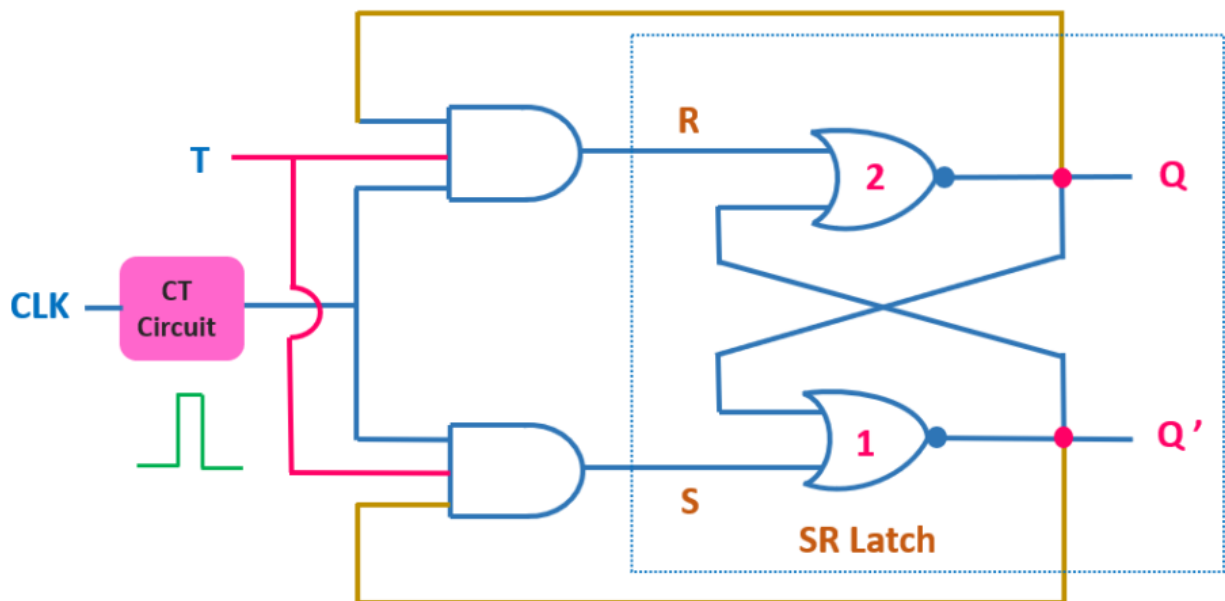
Truth Table

CLK	J	K	Q_{n+1}
↑	0	0	Q_n
↑	0	1	0
↑	1	0	1
↑	1	1	Q_n'

As you can see from the truth table, when both inputs of JK flip-flop are 0 then it holds the current state. And when its both inputs are 1, then the output of the flip-flop toggles. That means by connecting both inputs of the JK flip-flop together, it can be used as a T flip-flop.

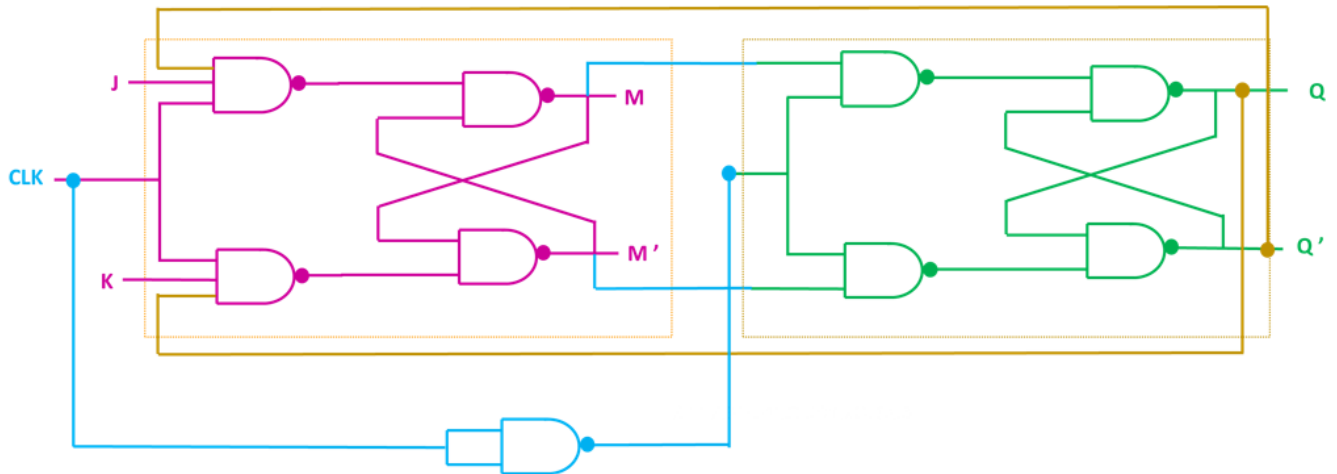


That means with the little modification in the circuit of the JK flip-flop, it can be used as a T flip-flop. The circuit of the T flip-flop is shown below.



Master Slave Flip-Flop

Circuit Diagram of Master Slave JK Flip-Flop:

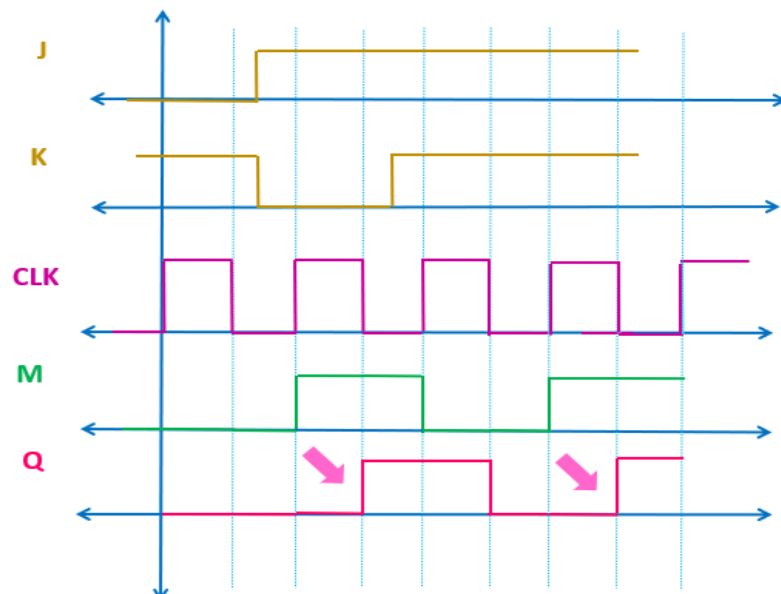


Circuit Diagram of Master Slave JK Flip-Flop

- As shown in the above figure, it consists of two gated SR latches.
- The first latch acts as a master latch and the second latch acts as a slave latch. The output of the master latch is connected to the slave latch.
- The Q' output of the slave latch is connected back to the master latch where the J input is applied and similarly, Q output is connected back where the K input is applied.
- The clock signal to the slave latch is applied through an inverter. That means when the clock signal is high, the master is enabled and the slave is disabled. And similarly, when the clock is low, the slave is active and the master is disabled.

Working of Master Slave JK Flip-Flop

Let's understand the working of Master Slave JK flip-flop using a timing diagram.



Timing Diagram of Master Slave JK Flip-Flop

Here, initially it has been assumed that the outputs of both master and slave are 0. (Both M and Q are 0). When the clock signal is high then master latch will get enabled and will respond to the input signals. In this case, initially during the first clock, $J = 0$ and $K = 1$. So, output of the master latch will be 0. During ON time of the clock, the slave latch will remain disabled and it will hold its current state. Similarly, during the OFF time of the clock, the master will get disabled and it will hold its current state. And at the same time, the slave latch will become active and will follow the master output. That means, the slave latch is following the master output during the off time of the clock. Or in other words, the slave latch follows the master output after the delay of T_{ON} . Where T_{ON} is the ON time of the clock signal. This cycle repeats at every clock cycle.

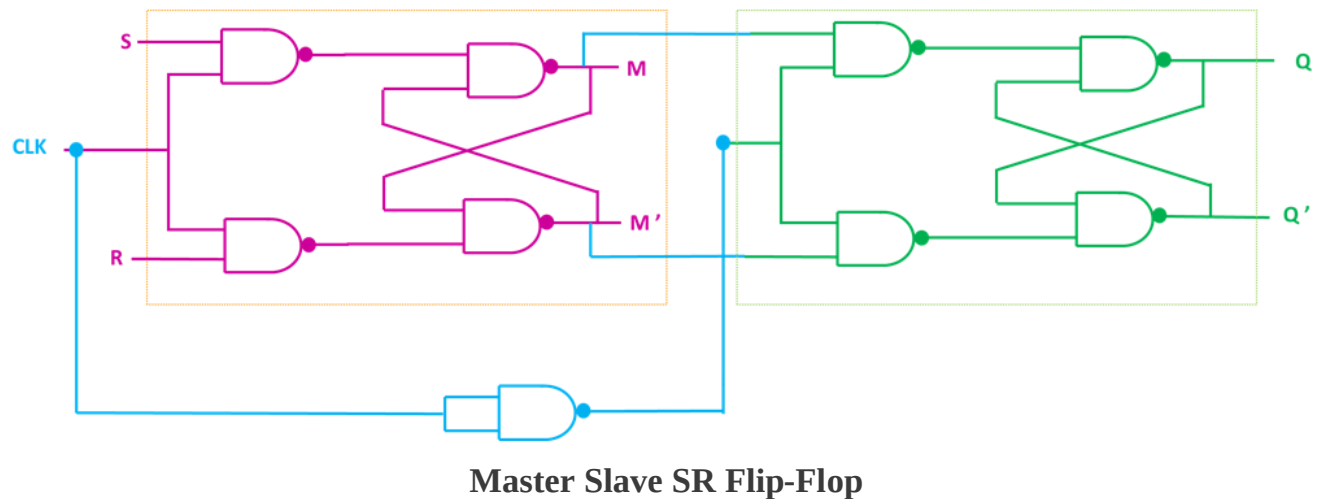
During the ON time of the second clock, since $J = 1$ and $K = 0$, so master output $M = 1$. And the slave output Q will remain 0. (Since it is disabled) The slave latch will follow the master output during the OFF time of the clock. From the timing diagram, you can see that, the slave is following the master output after the delay of T_{ON} .

For more information, please go through this video on the master slave flip-flop.

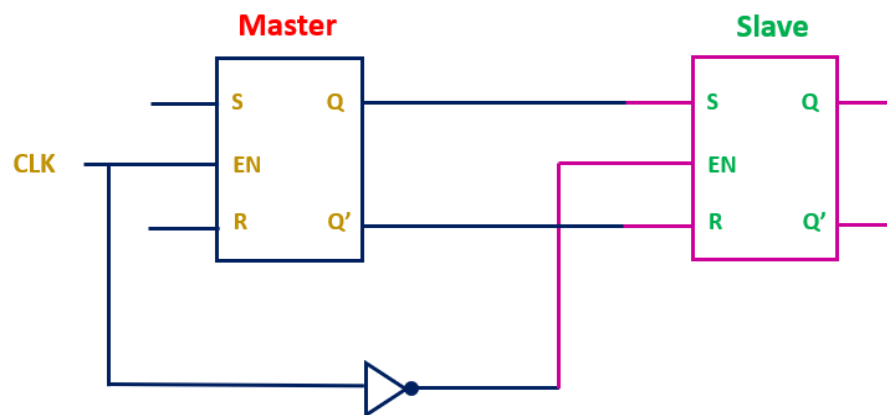
Master Slave SR Flip-Flop and D Flip-Flop

Similar to the master slave JK flip-flop, the master slave D flip-flop and SR flip-flop can be designed.

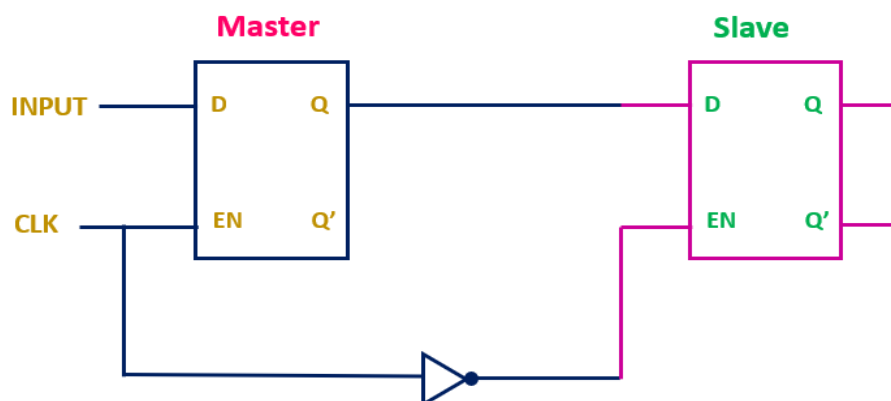
The circuit diagram of the master slave SR flip-flop is shown below.



In a simplified manner, this is how it can be represented.

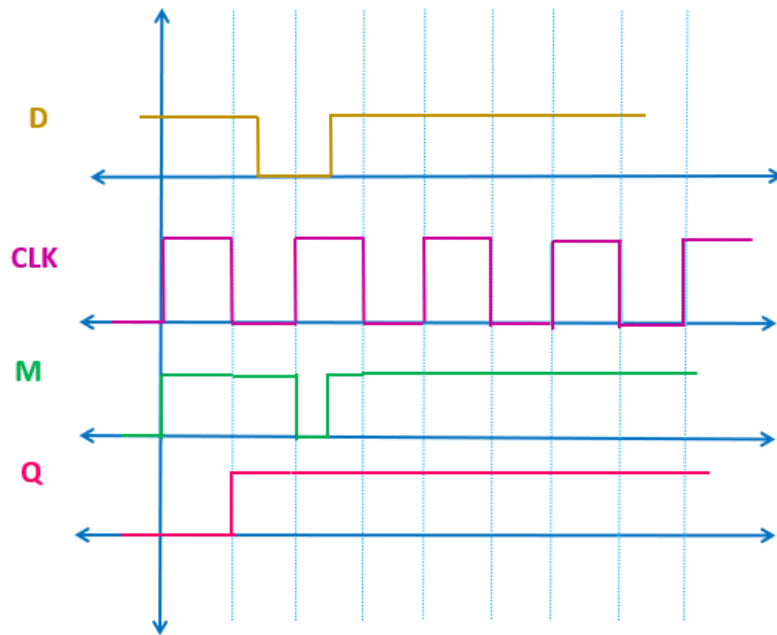


Similarly, using two gated D latches, the master slave D flip-flop can be designed.



Master Slave D Flip-Flop

The master slave flip-flop circuit works correctly when the input is constant during the ON time of the clock. If the input signal changes during the ON time of the clock, then slave will not be able to follow the master output. The same is shown in the below timing diagram.



As you can see in the timing diagram, during the second clock cycle, the D input is changing during the ON time of the clock. Since the master is ON at that time, so master will follow the change in the input signal. But the slave will not follow that input change. Because the slave is following the master output during the OFF time of the clock. Just at the end of the ON period of the second clock, the master output M is 1, so slave will follow the same output. That means, the slave is not able to follow the master output completely when the input is changing during the ON time of the clock.

That means, to use the master slave flip-flop in a correct manner, we need to ensure that the input is not changing during the ON time of the clock.